Kurt Matthew Nudo

krtmtthw.dev | github.com/abrycta

WORK EXPERIENCE

IBM **Feb 2026 – Present**

Experienced Full Stack Developer, Japanese Bilingual *Quezon City, Metro Manila (Hybrid)*

- Build backend modules in Java 17 and Jakarta EE, with MyBatis over Oracle and Maven builds, for a Japanese banking system migration. Have contributed to 20+ of its modules.
- Write and maintain 50+ JUnit and Mockito unit tests at 100% coverage, each with its own test data and database setup, for exception handling, business rules and database interactions.
- Have written ~40 MyBatis mappers and SQL queries against ~30 Oracle tables.
- Trace defects and edge cases to their root cause in the code or the Oracle data; 20+ fixed so far.
- Carry features from specification review through implementation, CI builds, release checks and deployment.
- Draft modules from reviewed design documents with AI-assisted code generation, then review, debug and refine them.

  Technologies: Java 17, Jakarta EE, MyBatis, Oracle Database, SQL, Maven, Git, JUnit, Mockito, CI/CD

Advanced World Solutions **Jul 2024 – Feb 2026**

Full Stack Developer *Makati, Metro Manila*

- Migrated 40+ of the project's 120 production modules (20+ in the first year) from a legacy C# desktop system to React, TypeScript and Node.js for a Japanese enterprise client.
- Cut a monthly report query from 6.2 s to 1.1 s by adding a composite index and rewriting the joins.
- Rebuilt 45 desktop screens as React pages and exposed 38 Oracle PL/SQL procedures through 30 REST endpoints for persistence and transactions.
- The team reached 96% feature parity against a 350-item checklist, on a modular codebase with async data handling.
- Wrote Jest unit and integration tests for those 40+ modules and kept their coverage at 92%+, above the 90% release bar.
- Closed 40+ verified bug tickets across UI workflows, API integrations and localized Japanese content.
- Worked directly with Japanese engineers to clarify specifications, settle ambiguous requirements and check the accuracy of the system's Japanese text.

  Technologies: TypeScript, JavaScript, React, Node.js, C#, Oracle PL/SQL, REST APIs, Git, Jest

PROJECTS

Lights **2026**

Web app: poverty estimates for 1,600+ Philippine cities and municipalities, 2012–2025 *lights.krtmtthw.dev*

- Builds on a co-authored Springer paper on forecasting poverty from nighttime lights and vegetation indices with recurrent neural networks in Python and TensorFlow (Lecture Notes in Networks and Systems, 2026, pp. 451–461).
- A Rust core reproduces a trained LightGBM model to within 3.7e-14 and runs it in the browser as WebAssembly. The 2023 nowcast error is 12% lower than a carry-forward baseline.

  Technologies: Rust, WebAssembly, LightGBM, Python, Google Earth Engine, React

Replay **2026**

Web app: typhoons that hit the Philippines or Japan since 1940 *replay.krtmtthw.dev*

- One OpenAPI 3.1 contract served by four backends (Spring Boot, Go, Node, Rust) that pass the same contract tests.

  Technologies: Java 17, Spring Boot, Go, Node.js, Rust, OpenAPI 3.1, PostgreSQL/PostGIS, React, TypeScript

SKILLS

Languages and frameworks: Java 17, Jakarta EE (Java EE), Spring Boot, MyBatis, TypeScript, JavaScript, React, Node.js, REST APIs, Python, Rust, SQL (Oracle, PostgreSQL/PostGIS)

Testing and delivery: Git, JUnit, Mockito, Jest, Maven, unit and integration testing, CI/CD, GitHub Actions

Certifications: PhilNITS Fundamental IT Engineers (FE), Dec 2024

Spoken languages: English; Japanese (JLPT N4, Jan 2025; studying for N3)

EDUCATION

Saint Louis University **Jul 2024**

Bachelor of Science in Computer Science, Summa Cum Laude (general average 93) *Baguio, Benguet*